

Ittyavira C Abraham

AI/ML Engineer | Machine Learning | Generative AI | FastAPI | Python

📍 Kottayam, Kerala, India

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🔗 [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience developing production-oriented AI applications through industry internships. Skilled in Machine Learning, Deep Learning, Generative AI, Retrieval-Augmented Generation (RAG), NLP, Python, FastAPI, Docker, SQL, and cloud platforms including Azure and GCP. Passionate about designing scalable AI systems and solving real-world problems through data-driven solutions.

SKILLS

Programming:

Python, SQL

AI/ML/DL:

TensorFlow, PyTorch, Scikit-learn, XGBoost, Neural Networks, OpenCV

Generative AI & NLP:

Transformers, Hugging Face, LangChain, RAG, BERT, GPT, Prompt Engineering, Vector Search

Data Science:

EDA, Feature Engineering, Statistical Analysis, Time Series Forecasting, Data Visualization

Cloud & Deployment:

Docker, Azure, GCP, Model Deployment

Databases:

PostgreSQL, MongoDB

Backend:

FastAPI, Django, Flask

Version Control:

Git, GitHub

Web:

HTML, CSS, SASS, JavaScript, React

PROJECTS

🔗 Predicting Safety Stocks using Machine Learning

- Developed a time-series forecasting model using XGBoost, SARIMA, and LSTM to predict safety stock levels.
- Used historical demand and lead-time data to optimize inventory levels and reduce stockouts.
- Improved prediction accuracy by 12% compared to baseline regression model.

🔗 BERT-based News Classifier

- Built a deep learning-based BERT model for multi-class news classification.
- Preprocessed and tokenized text using Hugging Face Transformers.
- Fine-tuned the model and evaluated performance using F1-score and confusion matrix.

🔗 AI RAG Assistant using LangChain

- Built a Retrieval-Augmented Generation (RAG) application using LangChain and Hugging Face embeddings.
- Implemented document-based question answering with semantic retrieval and context-aware responses.
- Integrated PDF processing and vector search for efficient knowledge retrieval.

PROFESSIONAL EXPERIENCE

AI/ML Developer Intern – Cyber Intelligence Corps (CIC)

May 2026 – Present | Hybrid | Thiruvananthapuram, Kerala

- Contributed to a production AI system using computer vision, deep learning, OpenCV, and NVIDIA Jetson.
- Developed REST APIs using FastAPI and React dashboard components for real-time monitoring.
- Implemented facial embeddings, FAISS vector search, Docker containerization and deployment, and version-controlled workflows.

AI/ML Intern – Edunet Foundations (AICTE Internship)

Apr 2025 – May 2025 | Remote

- Trained and evaluated classification models using Azure Machine Learning Studio as part of an AICTE-Microsoft collaborative AI Program.
- Explored Azure AI Services including automated ML and model evaluation pipelines for supervised learning tasks.
- Completed hands-on modules covering end-to-end ML workflows – data preparation, model training, and performance evaluation.

EDUCATION

Master of Computer Applications (Artificial Intelligence)

Amrita Vishwa Vidyapeetham (Amrita Ahead), Coimbatore, Tamil Nadu, 2023 – 2025

Bachelor of Computer Applications

IFIM College, Bengaluru, Karnataka, 2019 – 2023,

CERTIFICATIONS

- **Microsoft AI & ML Engineering**  – (Coursera Professional Certificate) - 2025
- **IBM Generative AI Engineering with LLMs**  – (Coursera Professional Certificate) – 2025

PUBLICATIONS

- **"Role of Data Mining in Customer Relationship Management: A Conceptual Review"**  - International Journal of Research (IJR), Vol. 9, Issue 04, April 2022